MILITARY SPECIFICATION

WARHEAD SECTION, TRAINING: M74, PACKING AND HANDLING

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for the packing and handling of the Nike Hercules M74 training warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8846000	- Warhead Section, Forward Body Section, Atomic, Training: M74.
8846004	- Carton, Static Tube Self-aligning.
8846005	- Carton, Static Tube Self-aligning.
8846373	- Warhead Section Assembly - Packaging Drawing.
EL-8855813	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The trainer warhead shall comply with all requirements specified on the drawing (dwg) 8846000, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Environmental testing (see 4.4.1).

3.5.1 Shock. The trainer warhead section shall withstand drop tests while enclosed in its case without evidence of physical damage or impairment of operation when tested as specified in 4.4.1.1.

3.5.2 Vibration. The trainer warhead section shall be vibrated at the frequencies, acceleration and number of sweeps listed in Table I in the vertical plane. Upon completion of the vibration tests the warhead section shall show no evidence of physical damage or impairement of operation when tested as specified in 4.4.1.2.

Table I
Vibration

Cycles of frequency Range (Hz)	Sweep Duration (Minutes)	Constant Acceleration or Constant Double Amplitude	Number of Sweeps
5-30-5	6 ± 0.5	0.06 ± 10 percent	1
30-500-30	10 ± 0.5	10 g's ± 10 percent	1

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.6.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

3.6.7 Stitching. Stitching shall not be missing where required. Stitching shall be free from any breaks or skipped areas. Tight stitching which breaks when the material is pulled to a flat surface or specified contour, shall be rejected.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8846000) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

2.	Shock (see 3.5.1)	X
3.	Vibration (see 3.5.2)	X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 *Carton, prior to sealing, (see dwg 8846373).*

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Screw and washer missing	Visual
	102. Band assembly missing	Visual
	103. Threads on plug adaptor damaged to extent that it prevents subsequent assembly	Visual
	104. Plug assembly (safe, Arm, S to A and S to S) missing	Visual
	105. Shorting plug and bracket assembly, missing or damaged	Visual
	106. Cable clamps, missing or damaged	Visual
Minor:	None defined.	

4.3.2.1.2 Carton sealed (see dwg 8846373).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Carton damaged to extent that components are exposed or can fall out	Visual
	102. Sealing strips missing or partially detached	Visual
Minor:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

4.3.2.1.3 Container M409, prior to closing (see dwg 8846373).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Warhead section missing	Visual
102.	Carton missing	Visual
103.	Disiccant bags missing, torn or insufficient	Visual
104.	Foreign matter in container	Visual
105.	Weld cracked	Visual
106.	Container dented or damaged to the extent it will not close or protect contents	Visual
107.	Bolts between warhead at station 0.150 and the thrust mount bracket missing	Visual
108.	Bolts between thrust mount and movable track missing	Visual
109.	Hoisting beam left in place	Visual
110.	Movable track clamp at station 87.5 missing	Visual
111.	Plastic pluts in lifting eye socket missing	Visual
Minor:	None defined.	

4.3.2.1.4 Container sealed (see dwg 8846373).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Markings missing, misleading, or unidentifiable	Visual
102.	Painting defective to extent it shows the primer coat or bare metal	Visual
103.	Closure lifting handles missing, or damaged	Visual
104.	Latching handles missing or inoperative	Visual
105.	Latching handles wrench missing	Visual
106.	Humidity indicator card missing	Visual
107.	Humidity indicator shows pink at "30"	Visual
108.	Glass over humidity indicator broken or missing	Visual
109.	Cover over humidity indicator missing	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

Minor: None defined.

4.3.2.1.5 Static tube self aligning, carton, prior to sealing (see dwgs 8846004 and 8846005).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Static tube missing	Visual
	102. Static tube not in protector assembly	Visual
	103. Static tube loose in protector assembly (nut loose)	Visual
	104. Static tube protector assembly missing from carton	Visual
	105. Carton damaged to the extent that contents can be damaged	Visual
	106. Foreign material in carton	Visual
Minor:	None defined.	

4.3.2.1.6 Static tube self aligning carton sealed.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Carton damaged to extent that components are exposed or can fall out	Visual
	102. Sealing strip missing or partially detached	Visual
	103. Marking missing, misleading or unidentifiable	Visual
Minor:	None defined.	

4.3.2.1.7 Nose section assembly crate, prior to being closed in (see dwg 8846374).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Nose section assembly missing	Visual
	102. Static tube carton missing	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

103. Bolts that secured nose section assembly to base loose or missing.....Visual
104. Metal strapping missing, or loose.....Visual
105. Wood intermediate support missing, cracked or not secured.....Visual
106. Felt pads on intermediate support missing or partially detached.....Visual

Minor: None defined.

4.3.2.1.8 Nose section assembly crates closed up.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Markings missing, misleading or unidentifiable.....Visual
102. Boards or plywood split or broken.....Visual
103. Nails protruding.....Visual
104. Shipping paper missing, torn or insecure.....Visual

Minor: None defined.

4.3.3 Testing. There are no production tests applicable to the trainer warhead.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8855813). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Environmental testing (see 3.5).

4.4.1.1 Shock.

4.4.1.1.1 Flat drop. The loaded container shall be raised and allowed to fall freely from a height of 12 inches to a concrete or similarly hard surface, landing flat on the skids. Upon completion of test the container shall be examined for any damage which would impair its intended use.

4.4.1.1.2 Rollover. The loaded container, erect on its skids on a hard level floor shall be tipped slowly sideways until it falls freely and by its own weight to the floor on its side. This process shall be repeated for one revolution with falls from side to top, from top to other side and from the other side to the base. Upon completion of test, the container shall be examined for any damage which would impair its intended use.

4.4.1.1.3 Rotational drop. The loaded container shall be suported at one end of its base on a sill approximately five inches in height and placed at right angles to the skids. The other end of the container shall be raised and allowed to fall freely from a 12 inch height to a concrete or similarly hard surface. This test shall be repeated at 24 and 36 inch increments. These tests shall be applied to both ends of the container. After each test the container shall be examined for any damage which would impair its intented use.

4.4.1.1.4 Edgewise drop. The loaded container shall be supported at one corner of its base on a block approximately five inches high and with a block approximately 12 inches high placed under the other corner of the same end of the container. The opposite end of the container shall be raised and allowed to fall freely from a 12 inch height, measured from the lower of the two corners, to a concrete or similarly hard surface. This test shall be repeated at 24 and 36 inch increments. These tests shall be applied to one corner of each end of the container. After each test the container shall be examined for any damage which would impair its intended use.

4.4.1.2 Vibration. The trainer warhead section in its shipping container shall be mounted to a suitable vibration producing apparatus and vibrated as specified in Table I in the veritcal plane. Upon completion of the test the shipping container shall be opened and the warhead section inspected for damage.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

5.1.1 Level A. The trainer warhead shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The trainer warhead shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Warhead section. The nose section assembly and its components shall be packed as per dwg 8846374 and in accordance with the minimum package and pack which will afford adequate protection against corrosion, deterioration, and physical damage during the shipment from supply source to the first receiving activity for immediate use. Shipping containers shall be snugly packed and constructed so as to insure acceptance by a common carrier or air express for safe transportation to the point of destination. Containers shall conform to the requirements of the consolidated Freight Classification rules in effect at the time of shipment, except that fiberboard, when used shall have a minimum Mullen Test of 275 pounds. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.3 Marking.

5.3.1 M-409 container. The M-409 container shall be marked in accordance with TM39 W31, 8-9 as specified for training use.

5.3.2 Crate. The crate for the nose section assembly and other components (see 5.2.2) shall be marked in accordance with dwg 8846374.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-2124 dated 14 March 1962.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A058

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:41Z
Check the source to verify that this is the current version before use.